Dancing Hexapod

Yecheng Zheng, Daniel Lazaro, Hongfei Liu, Hongru Liu

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We are building a assistive mobility hexapod for AXIS Dance Company

Robot used in this case is from 2025 Mechatronic Design team

Design requirements

$1000 budget for assembly
$2500 budget for motors, drivers, encoder, battery

Weigh less than 150 lb, while carrying more than 170lb
No discernible damage when dropping 25 lb loads on top

Legs must drive locomotion (aux. Wheels allowed support only, no power/odom)

Ability to move forward, backward, and turn either direction with zero radius

At least 30 min battery life in worst-case motion

Maximum: 30 seconds for 10’ forward motion
45 seconds for 10’ forward curvilinear - unknown angle!
45 seconds for 10’ backward motion
60 seconds for 10’ backward curvilinear - unknown angle!

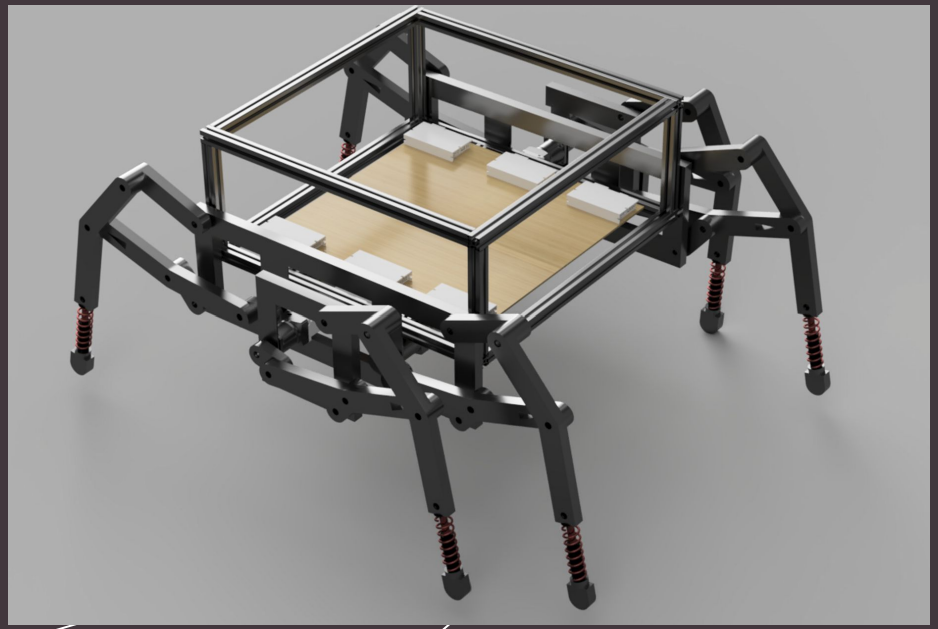
Goal: Be faster than the other (3) mechatronic teams!

We successfully beat all other hexapod teams

(since no one else's moved) - ""Final"" Demo 1 Today (Time Optimal Control)

We pivoted to assemblies with cnc " knees", press fitted and JB welded together.

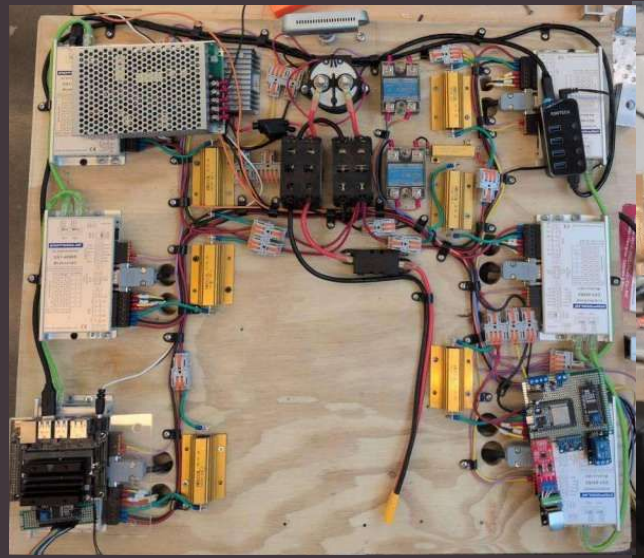
We optimized our budget by using amazon returns to test out spring-mass dampers, but realized these were all too stiff.

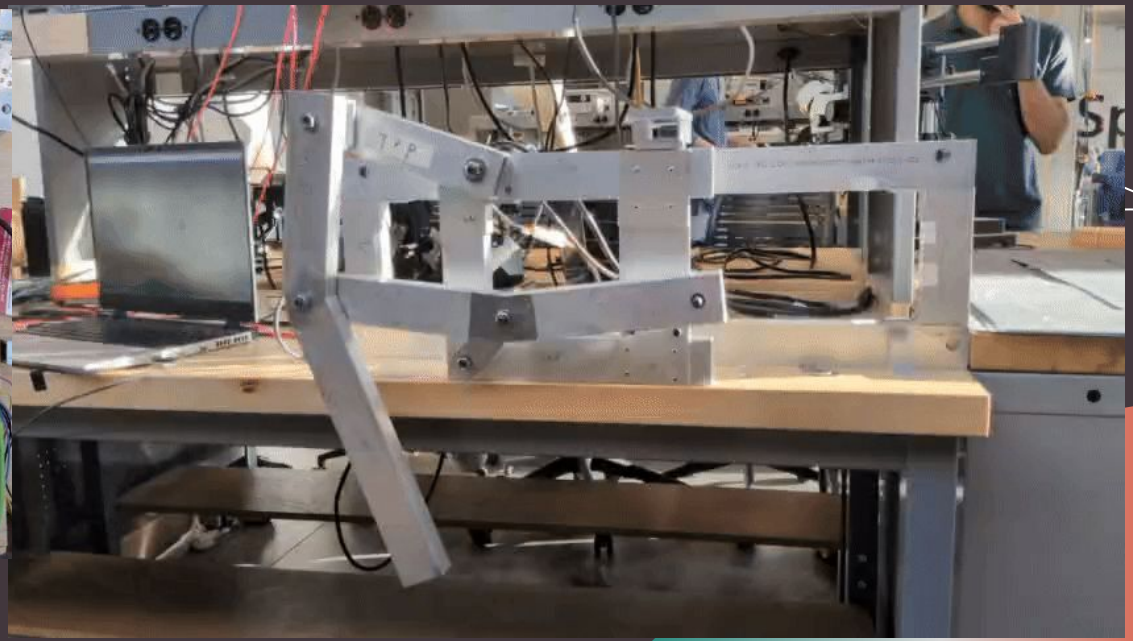


OLD

NEW

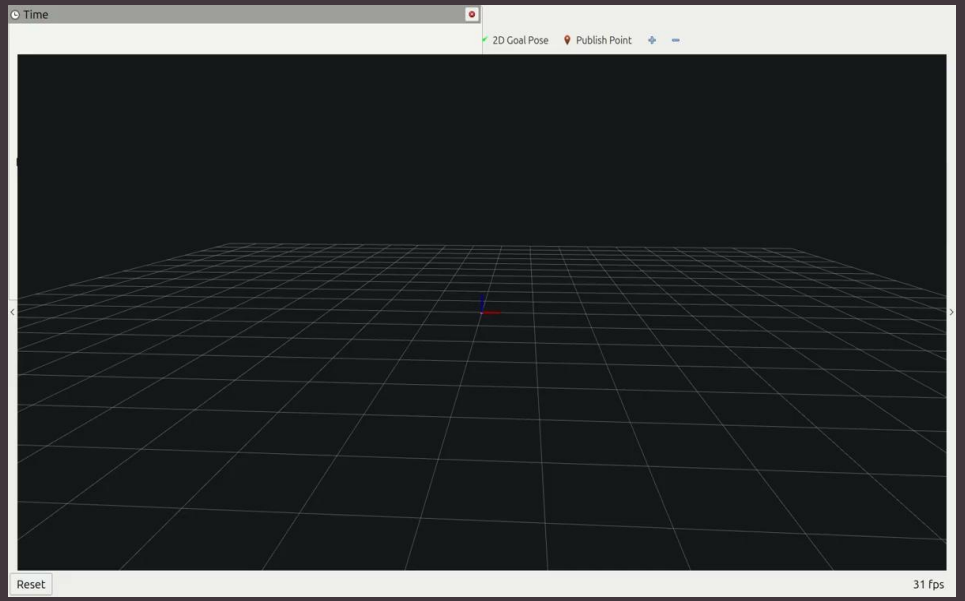
The planned electrical design successfully was able to reliably power motors, but introduced communication constraints.





Next steps - ""Final"" Demo 2 (Apr 22)

1. tune each physical leg height and linkage spacing
2. test gait controller on ESP32
3. mathematically model forward kinematics for each leg
4. update mujoco model to resemble final assembly as closely as possible
5. update mujoco controller to work with physical motor communication constraints, re-run MJX rollout optimizing
6. translate servo encoder to body odometry on ESP32
7. add imu feedback to controller on ESP32
8. mount and test foot touch sensing to correct odometry error

Next steps - Public Demo (Apr 29)

Stress test battery capacity by walking at 0.33 ft/s forward velocity
test curvilinear, forward/back path following at several arc angles.
tune imu / encoder bias to reduce (likely large) odom error.
run tests with and without human rider.
Test gait control with jetson
lightweight planning informed by mapping of area.